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Visual Content Captioning and Audio Conversion Using CNN-RNN with Attention Model

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ABSTRACT

The primary objective of this research is to develop an image captioning and audio conversion system based on Convolutional Neural Networks (CNN) and Recurrent Neural Networks (RNN) with the integration of an Attention Mechanism, aimed at improving accessibility for visually impaired individuals. The research design follows a systematic approach involving data collection, preprocessing, model development, training, evaluation, and implementation. The methodology utilizes CNN for visual feature extraction, RNN for language modeling, and an Attention Mechanism to enhance contextual relevance in caption generation. Google Text-to-Speech (gTTS) is also integrated to convert generated captions into audio format. The main outcomes demonstrate that the model is capable of generating coherent and contextually relevant captions, as validated through qualitative assessment and quantitative measurement using the BLEU score. Experimental results show decreasing training and validation loss over 8 epochs without signs of overfitting, indicating stable model performance. The attention visualization confirms the model’s ability to focus on relevant image regions during caption generation. In conclusion, the proposed CNN-RNN architecture with Attention effectively generates descriptive captions and converts them into speech, showing strong potential for real-world accessibility applications.

1. INTRODUCTION

The development of technology in the fields of image and audio processing has progressed rapidly, opening up significant opportunities to create applications that provide convenience for human life. One technology that is increasingly gaining interest is visual content captioning, which is the process of generating textual descriptions of visual content such as images, which can then be converted into audio form [1]. This technology holds immense potential, particularly for visually impaired individuals who rely on audio-based systems to interpret their surroundings. However, achieving accurate and efficient conversion of visual content into meaningful spoken language remains a significant challenge due to factors such as varying lighting conditions, complex backgrounds, and unclear object boundaries.

The main problem in this research is the difficulty of accessing visual information for visually impaired individuals, as well as the limitations of image-to-speech conversion systems in generating accurate and contextual descriptions without being affected by environmental conditions such as lighting, complex backgrounds, or low image quality. Moreover, many existing systems still rely on intermediate text-based processes, which increase complexity and the potential for errors. Developing a system that

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can directly and efficiently translate visual content into understandable speech remains a significant challenge in the fields of image processing and pattern recognition.

Recent studies have explored direct image-to-speech conversion using various approaches, including an end-to-end method without text supervision that employs a VQ-VAE to learn discrete speech representations [2], integrated image preprocessing to enhance OCR accuracy before audio conversion [3], and the use of OCR and TTS techniques in a machine learning-based system specifically designed for visually impaired users [4]. Other research highlights advancements in multimodal synthesis and interaction among images, text, and speech, where approaches such as transformers and adversarial learning have enabled more natural and contextual audio descriptions [5]. Additionally, systems based on NLP and GUI have been developed to facilitate low-cost and user-friendly access to visual information through audio [6], while another solution reported a high accuracy of up to 96.60% in image-to-speech conversion, enhancing independence and confidence among visually impaired individuals [7]. Overall, these approaches demonstrate significant progress in developing effective and practically applicable image-to-speech systems.

To address the challenge of converting visual content into speech, this research employs a CNN-RNN architecture enhanced with an Attention Mechanism to develop an image-to-speech system. CNN is used to extract meaningful features from images, enabling accurate object and scene recognition [8][9]. These visual features are then passed to an RNN, which generates descriptive text sequences based on the extracted image content. The addition of the Attention Mechanism allows the model to dynamically focus on relevant regions of the image when generating descriptions, improving the accuracy and contextual relevance of the output [10]. Finally, the generated text is converted into speech using Text-to-Speech (TTS) technology, creating a complete image-to-speech pipeline. The system is trained using image datasets from Kaggle and aims to enhance accessibility for visually impaired individuals by transforming visual information into spoken words effectively and efficiently [11]. The primary objective of this research is to develop an automated visual content captioning and audio conversion system that enhances accessibility for individuals with visual impairments. By integrating deep learning models with attention-based mechanisms and speech synthesis, this study aims to provide an inclusive, efficient, and accurate solution for translating visual information into spoken language. The scope focuses specifically on image-to-text captioning followed by text-to-speech conversion, using publicly available datasets for training and evaluation.

2. METHOD

The research stages in Figure 1 are designed to produce a Convolutional Neural Network (CNN)-based model capable of performing visual content captioning and audio conversion automatically by utilizing image and text data.

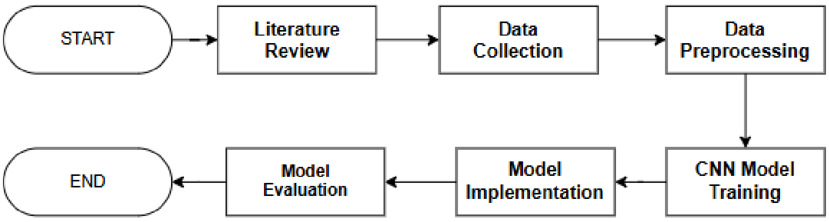

Figure 1. Research Stages

The method used in this research combines CNN and Recurrent Neural Network (RNN) with the support of an Attention Mechanism, which enables the system to focus on important parts of the image while generating descriptions [12]. The following are the stages conducted in this research:

1. **Literature Review**

A comprehensive review of relevant studies was conducted to establish a theoretical foundation for building an effective image-to-speech system. This includes prior works on CNN-based feature extraction, RNN-based caption generation, and attention mechanisms used in multimodal learning tasks [13]. This stage supports the selection of suitable architectures and methodologies for implementation.

2. **Data Collection**

Image datasets accompanied by textual captions were collected from Kaggle, which provides diverse and labeled data suitable for training deep learning models in image captioning tasks [14]. These datasets are essential for teaching the model to recognize visual patterns and associate them with appropriate descriptive text.

3. **Data Preprocessing**

Prior to model training, both image and text data underwent preprocessing. For images, resizing, normalization, and conversion into numerical arrays were performed to match input requirements for CNN processing [15]. For text, tokenization, sequence padding, and vocabulary creation were carried out to prepare captions for RNN training. Special tokens such as **<start>** and **<end>** were added to guide sentence generation [16] [17].

4. **Model Training**

A CNN-based encoder was implemented to extract high-level features from images. The AlexNet architecture was selected due to its effectiveness in hierarchical pattern recognition within two-dimensional image data [18][19]. Feature extraction was followed by convolutional, pooling, and activation layers to produce robust visual representations [20].

5. **Model Implementation**

Extracted image features were fed into an RNN-based decoder equipped with an Attention Mechanism. The decoder generated descriptive sentences based on the visual content, while the attention module enabled dynamic focus on relevant image regions during word prediction [22] [23]. Finally, the generated captions were converted into speech using Google Text-to-Speech (gTTS), completing the image-to-speech pipeline.

6. **Model Evaluation**

The performance of the system was evaluated using greedy search and beam search methods for caption generation. The BLEU score was employed to quantitatively assess the similarity between generated captions and reference texts [24]. Additionally, qualitative evaluation focused on the contextual accuracy of the generated descriptions and the clarity of the resulting audio output after TTS conversion.

3. RESULTS AND DISCUSSION

3.1. Data Understanding

The research begins with the data understanding stage, where a labeled image dataset along with corresponding textual captions was imported from Kaggle specifically the Flickr8k dataset , which contains 8,091 images and 40,455 descriptive captions . Each image typically depicts common objects, people, or scenes in real-life contexts, such as people walking, vehicles on roads, or animals in natural environments. The captions describe the visual content of each image using simple English sentences.

```
print("Total captions present in the dataset: "+ str(len(annotations)))
print("Total images present in the dataset: " + str(len(all_img_path)))

Total captions present in the dataset: 40455
Total images present in the dataset: 8091
```

Figure 2. Data Understanding Results

Figure 3 illustrates the initial data structure, showing how the dataset was loaded into two separate variables: one containing file paths to images and another containing their corresponding captions. This separation allows for parallel processing of visual and textual modalities during later stages.

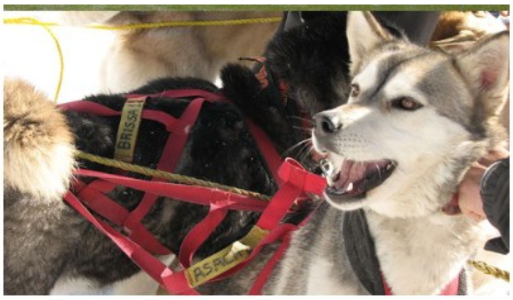

Figure 3. Image Dataset Visualization

Figure 4 displays sample images from the dataset, providing a visual overview of the types of scenes included ranging from outdoor landscapes to indoor settings. These images vary in lighting conditions, object positions, and background complexity, making them suitable for testing robustness in caption generation.

	ID	Captions	Path
0	1000268201_693b08cb0e.jpg	A child in a pink dress is climbing up a set of stairs in an entry way .	/content/drive/MyDrive/Flickr/Dataset/Flicker8k_Dataset/1000268201_693b08cb0e.jpg
1	1000268201_693b08cb0e.jpg	A girl going into a wooden building .	/content/drive/MyDrive/Flickr/Dataset/Flicker8k_Dataset/1000268201_693b08cb0e.jpg
2	1000268201_693b08cb0e.jpg	A little girl climbing into a wooden playhouse .	/content/drive/MyDrive/Flickr/Dataset/Flicker8k_Dataset/1000268201_693b08cb0e.jpg
3	1000268201_693b08cb0e.jpg	A little girl climbing the stairs to her playhouse .	/content/drive/MyDrive/Flickr/Dataset/Flicker8k_Dataset/1000268201_693b08cb0e.jpg
4	1000268201_693b08cb0e.jpg	A little girl in a pink dress going into a wooden cabin .	/content/drive/MyDrive/Flickr/Dataset/Flicker8k_Dataset/1000268201_693b08cb0e.jpg

Figure 4. Textual Dataset Visualization

Figure 5 presents a selection of raw caption texts associated with the images. The captions are short, single-sentence descriptions written by human annotators. Examples include "a woman walks down a sidewalk" or "a dog is playing in the grass." These captions serve as the ground truth for training and evaluating the model's ability to generate accurate spoken descriptions.

3.2. Text Preprocessing

During text preprocessing, the captions were tokenized to build a vocabulary of up to 5,000 unique words. Any word outside this limit was replaced with the special token **<unk>**. Each word was then mapped to a numerical index to enable model training. To mark sentence boundaries, special tokens **<start>** and **<end>** were added to each caption.

```
0                 <start> A child in a pink dress is climbing up a set of stairs in an entry way . <end>
1                                                   <start> A girl going into a wooden building . <end>
2                                         <start> A little girl climbing into a wooden playhouse . <end>
3                                     <start> A little girl climbing the stairs to her playhouse . <end>
4                                <start> A little girl in a pink dress going into a wooden cabin . <end>
5                                              <start> A black dog and a spotted dog are fighting <end>
6                   <start> A black dog and a tri-colored dog playing with each other on the road . <end>
7    <start> A black dog and a white dog with brown spots are staring at each other in the street . <end>
8                         <start> Two dogs of different breeds looking at each other on the road . <end>
9                                       <start> Two dogs on pavement moving toward each other . <end>
Name: Captions, dtype: object
```

Figure 5. Text Preprocessing Results

Figure 6 shows the frequency distribution of the top 30 most-used words before and after preprocessing, highlighting the dominance of common nouns, verbs, and prepositions in the dataset. Figure 7 visualizes the tokenizer output, demonstrating how sentences are transformed into sequences of integers ready for model input.

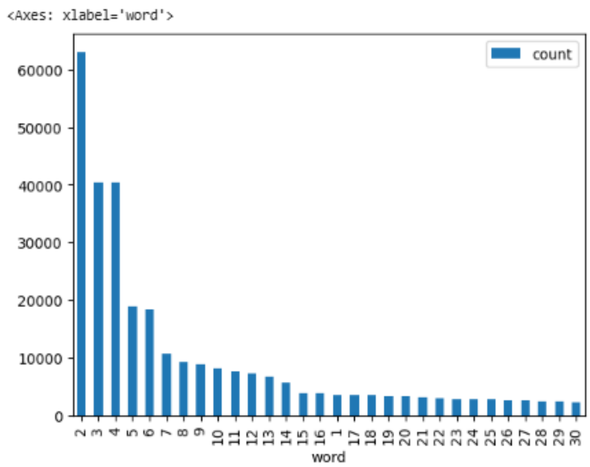

Figure 6. Text Tokenizer Visualization

3.3. Image Preprocessing

All images were resized to 299x299 pixels and normalized according to the preprocessing requirements of the InceptionV3 model a pretrained CNN trained on the ImageNet dataset. This ensures compatibility when extracting high-level visual features.

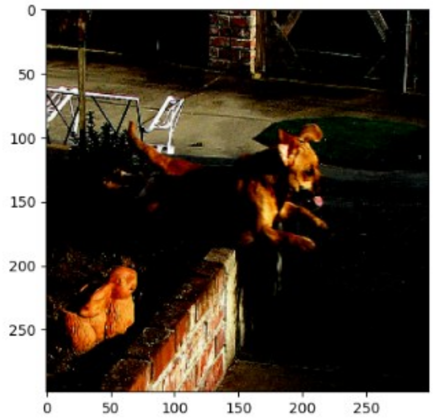

Figure 7. Image Preprocessing Results

Image features were extracted from the final layer of InceptionV3, producing a feature map of shape 8x8x2048 per image. These features were stored in NumPy arrays and dictionaries for efficient access during training.

3.4. Dataset Splitting

The dataset was split into training (80%) and testing (20%) sets using a fixed random state (random_state=42) to ensure reproducibility. Both subsets contain pairs of image features and processed captions.

```
Training data for images: 32364
Testing data for images: 8091
Training data for Captions: 32364
Testing data for Captions: 8091
```

Figure 8.Train and Test Dataset

Figure 9 shows the distribution of training and test data, while Figure 10 visualizes how the data is batched during training. Batching helps stabilize training performance and improves memory efficiency.

```
[ ] sample_img_batch, sample_cap_batch = next(iter(train_dataset))
    print(sample_img_batch.shape)  #(batch_size, 8*8, 2048)
    print(sample_cap_batch.shape) #(batch_size,max_len)

    (32, 64, 2048)
    (32, 39)
```

Figure 9. Dataset Batch Division

3.5. Model Development

The proposed system integrates three key components: a CNN-based encoder , an attention mechanism , and an RNN-based decoder . The encoder uses pre-trained InceptionV3 weights to extract visual features, while the decoder generates captions using a GRU-based architecture.

The attention mechanism enables the model to dynamically focus on specific regions of the image when predicting each word. For example, when generating the word "woman," the model focuses on the region containing the person in the image.

```
Feature shape from Encoder: (32, 64, 256)
Predcitions shape from Decoder: (32, 5001)
Attention weights shape from Decoder: (32, 64, 1)
```

Figure 10. Model Development

Figure 11 provides an overview of the complete model architecture, illustrating how these components interact to produce meaningful captions.

3.6. Model Training and Optimization

The model was trained using teacher forcing, where the true previous word is used as input to the decoder rather than the model's own prediction. This speeds up convergence and stabilizes learning.

Training was conducted over 9 epochs, with both training and validation losses decreasing steadily, as shown in Figure 12. The absence of overfitting is evident from the consistent decline in both loss curves without divergence. This indicates that the model generalizes well to unseen data.

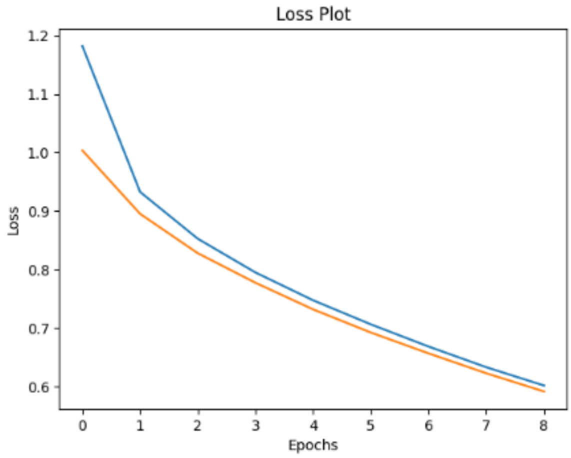

Figure 11. Model Training Visualization

Stability during training is also supported by:

1. Batch normalization layers in the encoder,
2. Dropout regularization in the decoder,
3. And early stopping based on validation loss monitoring.

3.7. Model Evaluation

Model evaluation was performed using greedy search decoding and BLEU score metrics. Greedy search selects the most probable word at each step, resulting in coherent and contextually relevant captions. Figure 13 shows an example prediction: "a woman walks down a sidewalk," which accurately describes the main subject and action in the image. The attention maps highlight relevant image regions for each generated word, confirming that the model focuses on appropriate visual cues.

Figure 12. Prediction Results for Image Captioning

A BLEU-2 score of approximately 23% was achieved, indicating moderate similarity between the predicted and reference captions. While not extremely high, this score reflects reasonable fluency and relevance in the generated text.

For the audio conversion stage, the predicted captions were converted into speech using Google Text-to-Speech (gTTS) . Figure 14 shows the caption used for audio generation, and the resulting audio file was played back successfully, confirming the feasibility of the end-to-end image-to-speech pipeline.

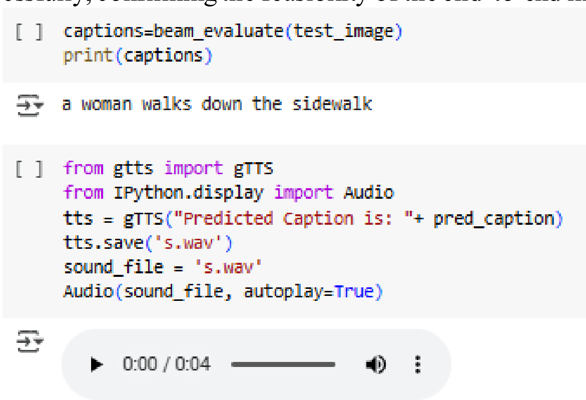

Figure 13. Image Prediction Audio Generation Results

4. CONCLUSION

In conclusion, this research successfully developed a visual content captioning and audio conversion system using a CNN-RNN architecture with an attention mechanism. The model demonstrated the ability to generate coherent and contextually relevant image descriptions, as evidenced by qualitative analysis of the predicted captions and supported by quantitative evaluation using the BLEU score. Although some predictions did not fully align with the actual image content, the attention mechanism effectively enabled the model to focus on key visual elements when generating captions. The integration of Google Text-to-Speech (gTTS) further extended the system's functionality by converting generated captions into audible speech, enhancing its potential for accessibility applications such as assisting visually impaired users. Despite minor semantic discrepancies in some outputs, the overall results affirm the effectiveness of combining CNN-based feature extraction, RNN-based language modeling, and attention mechanisms in building a robust image captioning system capable of real-world application. Looking ahead, future research can focus on improving the system's sustainability and scalability by exploring transformer-based models like Vision Transformers to enhance caption fluency and training efficiency. Expanding the dataset with more diverse scenes can improve generalization across various environments. Additionally, implementing end-to-end visual-to-speech mapping—without intermediate text—could reduce pipeline complexity. Finally, deploying the model on lightweight or mobile platforms may enable real-time assistive applications for visually impaired users.

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